

A NEW THEORY OF CREDIT LINES (WITH EVIDENCE)

Jason Roderick Donaldson

Naz Koont

Giorgia Piacentino

Victoria Vanasco

FACTS

Credit lines make up c. \$2T of committed credit, bulk of bank credit (80%)
Berg–Saunders–Steffen 20, Greenwald–Kraimer–Paul 21, Chodorow-Reich
et al. 21, and Sufi 09

Credit lines are rarely drawn (6%) even in crises (24%)
Ivashina–Scharfstein 10, Greenwald 21

Credit lines are bundled with loans, especially to risky firms (80%)
See below

Credit lines are sometimes revoked by lenders
Falato–Chodorow-Reich 22

QUESTIONS

Q1. Why are credit lines so common, even if rarely used?

Q2. Why are credit lines bundled with loans?

THIS PAPER

Dynamic model of borrower B issuing debt

Admati et al 17/DeMarzo–He 21

Friction: Non-exclusivity

After borrowing from one lender at t , borrows from another at $t + dt$

Innovation: Allow for credit lines (CLs)

Options to borrow $\tilde{p}d\tilde{Q}$ against face value $d\tilde{Q}$

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T. Empirical evidence: two natural experiments + cross-sectional tests

MODEL

OVERVIEW

Infinite-horizon sequential borrowing: B borrows from one lender at each t

Credit line-debt bundle at date 0 and new debt afterward

B's cost of debt $c(Q_t)$ increasing and concave in stock of debt Q_t

Captures expected coupon payment $\downarrow$ as default prob. $\uparrow$ in Q

Universal risk-neutrality, deep pockets, and discounting at rate ρ

NB: No revocation; no risk; no CLs at $t > 0$

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B's flow payoff: $v_t dt = y dt + p_t dQ_t - c(Q_t) dt$, where

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B's value: $V_t = \int_0^\infty e^{-\rho s} v_{t+s} ds$

LENDERS' VALUE

Lenders' flow payoff from unit debt given stock Q_t : $\gamma(Q_t)$

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Value of stream of coupons on unit of debt:

$$\Gamma_t = \int_0^\infty e^{-\rho s} \gamma(Q_{t+s}) ds$$

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Loans (p_t, dQ_t) : Borrow $p_t dQ_t$ against face value dQ_t

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($d\tilde{Q}$ put options on debt with strike $\tilde{p}$)

ASSUMPTIONS

A1: Dilution: $\gamma' < 0$ and $\lim_{Q \rightarrow \infty} \gamma(Q) = 0$

A2: Gains from trade:

Positive at $Q = 0$: $\gamma(0) > c'(0)$

Negative as $Q \rightarrow \infty$: $\gamma(\infty) < c'(\infty)$

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$$c(Q) = \underbrace{(1 - \pi(Q))\delta Q}_{E[\text{coupons}] = \gamma(Q)Q} - \underbrace{\tau_c \delta Q}_{\text{tax shield}} + \underbrace{\pi(Q)Qk}_{E[\text{default costs}]}$$

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Baseline assumptions easy to satisfy:

1. c increasing: $(\pi(Q)Q)' > 0$
2. c concave: $(\pi(Q)Q)'' < 0$
3. c' bounded above zero: $\lim_{Q \rightarrow \infty} (\pi(Q)Q)' > \bar{c}'_{lb} > 0$

SOLUTION CONCEPT: MARKOV PERFECT EQ.

At each date, lenders offer bundles/loans and B chooses one offer s.t.

B and lenders max expected future lifetime utility given beliefs

Beliefs are consistent

B's balance sheet $(Q_t, (\tilde{p}, d\tilde{Q}))$ is the state variable

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Notation: $\tilde{V}$ value function with credit line; V without (after draw down)

BENCHMARKS

BENCHMARK 1: EXCLUSIVE COMPETITION

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If B commits to one lender (of those competing at date 0) forever

B borrows at Date 0 and never again

Price is above marginal cost

EXCLUSIVE COMP: INTUITION

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Durable goods monopolist intuition: $MC = MR = (pQ)'$

Take effect of Q on p into account $\implies p > MC$

$\implies p > MC$: price higher/quantity lower than with perfect competition

BENCHMARK 2: NO CREDIT LINES (& NON-EXCLUSIVITY)

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Absent credit lines, with $dt \rightarrow 0$

B issues debt continuously

B captures no surplus

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Cf. Leverage ratchet effect and Coase Conjecture
(Admati et al. 18, DeMarzo–He 21, Coase 72,...)

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NB: c concave $\implies V$ convex ($V'' = -c''/\rho > 0$)

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Debt in place $\implies$ new debt tempting $\implies$ “ratchet effect” eats surplus

RESULTS

LEMMA: RATCHET EFFECT FOR CREDIT LINES

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Let B have CL $(\tilde{p}, d\tilde{Q})$ in place s.t. $\sim$ between drawing or staying at Q_0

B weakly prefers to draw for $Q > Q_0$

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Akin to leverage ratchet effect: Higher debt begets higher debt

R1: RATCHET-ANTI-RATCHET EFFECT

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Let $(\tilde{p}, d\tilde{Q})$ be CL s.t. $B \sim$ drawing at Q_0

B doesn't take any new debt ($dQ_t \equiv 0$)

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Step 3: Show $\underline{p} > \bar{p}$ for $d\tilde{Q}$ large enough

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NB: Self-enforcing independent of who holds debt

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Debt can be traded, as in practice, unlike in monitoring-based theories

R2: BUNDLING

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At date 0, B chooses a bundle of a loan (p, Q_0) and CL $(\tilde{p}, d\tilde{Q})$ s.t.

(p, Q_0) coincides with outcome of exclusive competition

$(\tilde{p}, d\tilde{Q})$ makes B indifferent to drawing at Q_0 with $d\tilde{Q}$ large

R2: BUNDLING: PROOF

Recall that under candidate bundle

B never borrows again by R1

B's payoff maximized s.t. lenders break even by BM1

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Can post candidate with price p replaced by $p + \epsilon$

R2: BUNDLING: INTUITION

Credit lines allow B to commit not to dilute by R1

Use credit lines to implement optimum without dilution (i.e. BM1)

REVOCATION

So far: Credit lines always honored if drawn

Now: Credit lines honored with prob. α (else disappear) if drawn

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$\implies$ Payoffs given drawing at Q_0 convex comb. of those at $Q_0 + d\tilde{Q}$ and Q_0

R3: LENDER REVOCATION RISK INCREASES B's DEBT

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With revocation risk $1 - \alpha$, B can sustain debt Q_0 to limit for $dt \rightarrow 0$:

$$\Gamma(Q_0) - \frac{c'(Q_0)}{\rho} \leq \frac{\alpha}{1 - \alpha} \frac{c'(\infty)}{\rho}$$

R3: REVOCATION RISK: INTUITION

Credit line revocation risk makes new lenders willing to lend at higher p

Less worried about dilution by credit line if might not be there

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R3: REVOCATION RISK: INTUITION

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Less worried about dilution by credit line if might not be there

$\implies$ harder to deter new debt

$\implies$ harder to commit not to dilute in the first place

PREDICTIONS & TESTS

PREDICTION 1: BUNDLING

Prediction: Credit lines are bundled with loans

Reason: CL protects loan from dilution

Evidence: > 70% term loans bundled with CL at origination

PREDICTION 2: BUNDLING PROPENSITY

Prediction: Bundling propensity higher if dilution risk higher

Reason: Protection valuable only if dilution likely

Evidence: More bundling if default likely or lenders less concentrated

PREDICTION 3: SIZE

Prediction: Credit lines are large

Reason: CL deters new borrowing only if large

Evidence: More bundling if default likely or lenders less concentrated

PREDICTION 3: SIZE

Prediction: Credit lines are large

Reason: CL deters new borrowing only if large

Evidence: More bundling if default likely or lenders less concentrated

PREDICTION 4: SIZE IN CROSS-SECTION

Prediction: Credit lines are larger if dilution risk higher

Reason: Deterrant more valuable when dilution more likely

Evidence: Larger CLs if default likely or lenders less concentrated

PREDICTION 5: DRAW DOWNS

Prediction: Credit lines are rarely drawn

Reason: CL off-equilibrium threat

Evidence: 45% of borrowers utilize none of CL; median 17%

PREDICTION 6: BORROWING

Prediction: Credit lines deter subsequent borrowing

Reason: Anti-ratchet

Evidence: Increased prob. CL present increased borrowing

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Prediction: Credit lines deter subsequent borrowing

Reason: Anti-ratchet

Evidence: Increased prob. CL present increased borrowing

Experiments: Changes in revocation prob. and regulatory cost of CL

PREDICTION 7: RATES

Prediction: Rates on CLs relatively low

Reason: Rate s.t. lender worse off when lends

Evidence: Median rate on CL $\approx$ that on the associated term loan

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In lit.: Insurance to take on more debt

Here: Commitment to take on less debt

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In lit.: Excessive debt and zero surplus

Here: Allowing for credit lines makes ratchet effect self-deterring

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Latent contracts:

In lit.: Help lenders support collusive outcomes

Here: Help borrowers support monopolistic outcome

CONCLUSION

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Heretofore unexplored role of credit lines: commit not to dilute

Curb competition with future self

Implement monopoly outcome